particles. Rather, they are tiny, one-dimensional filaments somewhat like infinitely thin rubber bands, vibrating to and fro. But don't let the name fool you: Unlike an ordinary piece of string, which is itself composed of molecules and atoms, the strings of string theory are purported to lie deeply within the heart of matter. The theory proposes that they are ultramicroscopic ingredients making up the particles out of which atoms themselves are made. The strings of string theory are so small—on average they are about as long as the Planck length—that they appear pointlike even when examined with our most powerful equipment.

Yet the simple replacement of point particles with strands of string as the fundamental ingredients of everything has far-reaching consequences. First and foremost, string theory appears to resolve the conflict between general relativity and quantum mechanics. As we shall see, the spatially extended nature of a string is the crucial new element allowing for a single harmonious framework incorporating both theories. Second, string theory provides a truly unified theory, since all matter and all forces are proposed to arise from one basic ingredient: oscillating strings. Finally, as discussed more fully in subsequent chapters, beyond these remarkable achievements, string theory once again radically changes our understanding of spacetime.1

A Brief History of String Theory

In 1968, a young theoretical physicist named Gabriele Veneziano was struggling to make sense of various experimentally observed properties of the strong nuclear force. Veneziano, then a research fellow at CERN, the European accelerator laboratory in Geneva, Switzerland, had worked on aspects of this problem for a number of years, until one day he came upon a striking revelation. Much to his surprise, he realized that an esoteric formula concocted for purely mathematical pursuits by the renowned Swiss mathematician Leonhard Euler some two hundred years earlier—the so-called Euler beta-function—seemed to describe numerous properties of strongly interacting particles in one fell swoop. Veneziano's observation provided a powerful mathematical encapsulation of many features of the strong force and it launched an intense flurry of research aimed at using Euler's beta-function, and various generalizations, to describe the surfeit of data being collected at various atom smashers around the